

Video Lecture Analysis Toolkit: An Open-Source Framework for Interactive Learning

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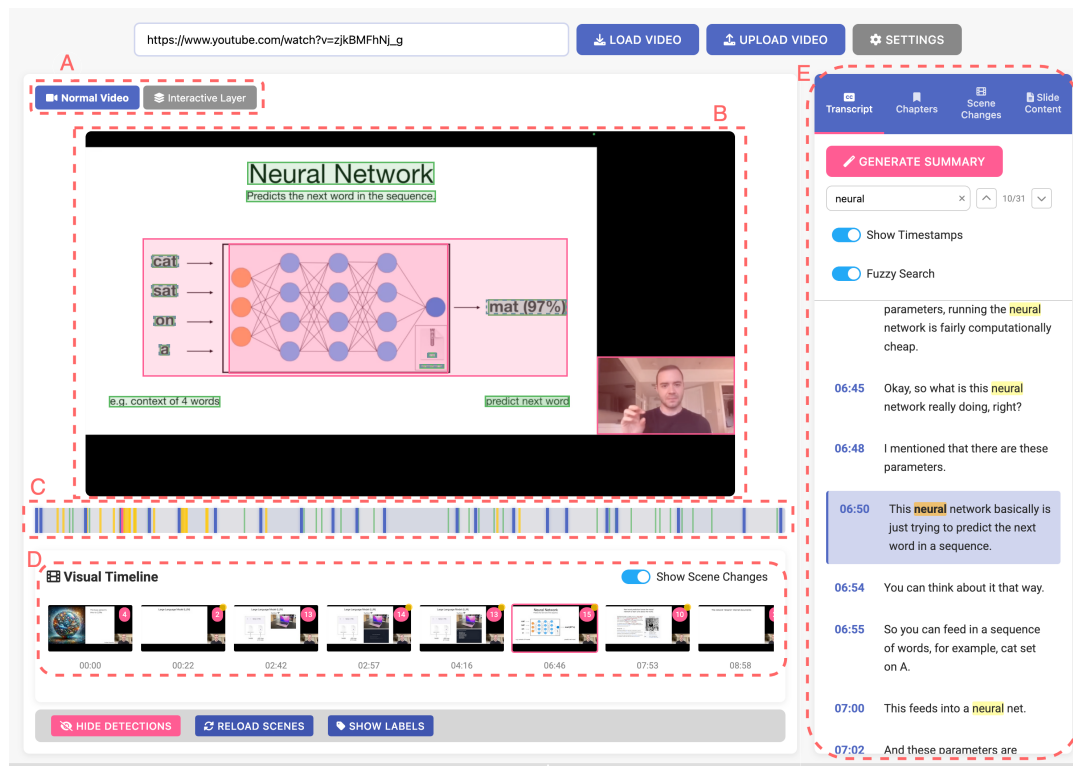


Figure 1: Interface of our application. (A) allows to user to turn on the interactive layer to move and resize elements in the video player (B). (C) represents the events timeline (slide transitions are marked in green, chapters are marked in blue) and search results in yellow. (D) represents a visual timeline with thumbnails for quick overview. (E) shows the transcript, chapters, scene changes and slide OCR content.

Abstract

The growth of online educational content, particularly slide-based video lectures, has created a need for tools that enhance navigation, comprehension, and accessibility. Many existing systems for video analysis are closed-source, hindering reproducibility and extension. To address this, we present the Lecture Video Analysis Toolkit, an

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open-source, proof-of-concept application designed for the multimodal analysis of slide video lectures. The toolkit integrates a processing pipeline that includes scene detection, visual entity extraction, transcription, optical character recognition (OCR), and semantic linking between spoken and visual content using embeddings. A key contribution is its interactive interface, motivated by early user feedback, that allows for customization of the viewing experience to suit individual preferences. The entire system is openly available and serves as a research prototype for validating the potential of multimodal analysis in creating more inclusive and improved learning experiences. A live demo is accessible at <https://travis-seng.fr/svla>, and the source code is openly available at <https://github.com/travisseng/svla-toolkit>.

CCS Concepts

• **Information systems** → **Multimedia information systems**; • **Computing methodologies** → *Computer vision*; *Natural language processing*.

Keywords

Open Source Software, Slide Video Lecture Analysis, Multimodal Learning, Accessibility, Interactivity, Scene Detection, Semantic Linking

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1 Introduction

Video lectures are a cornerstone of modern education, yet navigating their dense content remains a challenge. To address this, a body of research has produced systems to enhance the learning experience. Some focus on content extraction and organization through summarization and indexing tools [1, 6, 11–13]. Others have demonstrated the benefits of multi-layered interaction and adaptive layouts for retargeting content [5, 10].

While these systems offer valuable contributions and demonstrate the potential for enhancing educational videos, many remain closed-source and are difficult to evaluate, reproduce, or adapt in research contexts. To fill this gap, we introduce a proof-of-concept Slide Video Lecture Analysis Toolkit that integrates key capabilities—including scene detection, visual entity extraction, transcription, and cross-modal content linking—into a unified, open, and interactive application.

The design was motivated by early, informal feedback which highlighted the need for greater user control over the interface, leading to features that prioritize clear structure and customizable interfaces. The core contributions of our work are:

- **An Integrated Open-Source Pipeline:** We provide a fully open-source system combining multiple analysis modules into a single, reproducible application.
- **Multimodal Linking:** The system establishes semantic links between spoken words (transcript) and written text (OCR) using embeddings, allowing users to see the connection between what is said and what is shown.
- **Interactive UI:** The toolkit features an interactive user interface with an "Interactive Layer" that allows users to move, resize, or even hide video elements to reduce clutter and tailor the experience to their preferences.

This paper details the toolkit's architecture, its core processing modules, and the interactive interface designed to support inclusive education scenarios.

2 System architecture and pipeline

The application is built on a modular architecture comprising a backend processing pipeline, a file-based storage system, and a frontend web interface. The system is designed to be modular and reproducible, processing a one-hour video in approximately one

minute on a consumer-grade laptop with a NVIDIA RTX 4070 GPU. Figure 2 details the architecture.

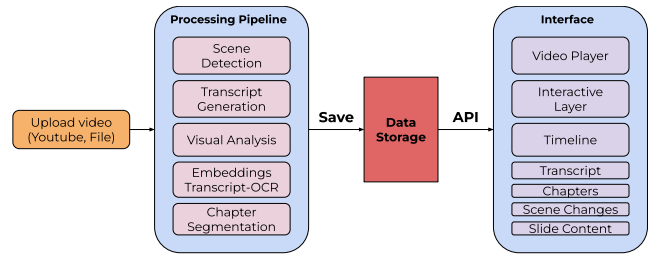


Figure 2: System architecture of our application designed around three components. The processing pipeline extracts multimodal information from the video then stores it in a file-based storage. A web interface queries data from the storage to display interactive information.

2.1 Processing Pipeline

The backend, built with **FastAPI**, orchestrates a sequence of analysis modules upon receiving a video from a user upload or a YouTube URL.

Video Ingestion: Videos are downloaded using **yt-dlp** or received via upload. A content-based hash is computed to uniquely identify each video and prevent redundant processing.

Scene Detection: The video is segmented into visually distinct intervals that typically align with slide transitions using *pyscenedetect* [2]. For each scene, a thumbnail is generated for the UI, and a full-resolution frame is saved for further analysis.

Transcript Generation: High-quality transcripts are generated using either the YouTube transcript API (when available) or OpenAI's Whisper model for offline processing.

Visual Content Analysis: A trained object detection model from [8] identifies elements of interest on each slide, such as titles, text paragraphs, diagrams, and images. Bounding boxes, labels, and confidence scores are saved for each detection. This lets us separate foreground from background, enabling accurate slide reconstruction.

Optical Character Recognition (OCR): Text regions identified by the object detection model are processed by an OCR engine. The toolkit supports both Tesseract [4] and the deep learning-based Surya OCR system [7]. The extracted text is aligned with its corresponding bounding box on the slide.

Multimodal Embedding and Linking: To connect the spoken and visual modalities, the system computes sentence-level embeddings for both the transcript and OCR text using the *all-mpnet-base-v2* model [9]. A cosine similarity score above a threshold of 0.5 is used to establish a semantic link between a spoken sentence and a piece of text on a slide if they belong to the same slide segment. This enables interactive cross-modal navigation.

Chapter Segmentation and Summarization: The application groups transcript segments into coherent chapters using a large language model (Gemini Flash 2.0 [3]) to provide a concise overview of the lecture's structure. This feature was motivated by user feedback emphasizing the need for clearer structural outlines.

3 User interface

The frontend, seen in Figure 1 is designed to be a clear and interactive dashboard for exploring the analyzed video content. It is divided into several panels that provide multiple, synchronized views of the lecture.

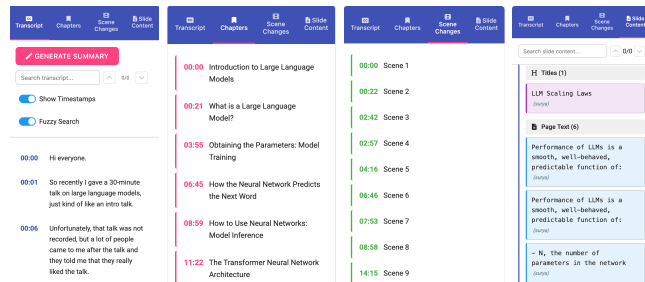


Figure 3: The interface contains a menu to display information. From left to right: the transcript menu, chapters menu, scene changes menu, and slide content menu.

The main interface components include:

Video Player and Timelines: A central video player is augmented by an event timeline below it, which marks slide transitions, chapter breaks, and search results. A visual timeline with clickable scene thumbnails provides a quick, linear overview of the presentation's visual flow.

Content Navigation Panel: A vertical tabbed panel on the right provides access to the full Transcript, a list of Chapters, a log of Scene Changes, and the OCR-extracted Slide Content, shown in Figure 3. All items in these tabs are timestamped and interactive; clicking any item seeks the video to the corresponding moment.

Integrated Search: A keyword-based search function allows users to locate exact matches across the spoken transcript and OCR-extracted visual text, providing a straightforward way to jump to specific terms or phrases.

Semantic Search and RAG Integration: Complementing keyword search, the toolkit supports semantic retrieval across transcripts, OCR-extracted text, and visual content (via image captioning). This enables retrieval-augmented generation (RAG) scenarios, where lecture material can serve as a knowledge base for tasks such as question answering and tutoring applications.

3.1 The Interactive Layer

A key innovation of our toolkit is the "Interactive Layer". When activated, this mode allows users to directly manipulate the visual components overlaid on the video player.

Content Customization: Users can move, resize, or completely hide any detected element, including the presenter's video, text boxes, or images. This feature was directly informed by feedback from neurodivergent users who expressed a desire to reduce visual clutter and cognitive load. For instance, a user can hide the presenter's face to focus solely on the slides, or enlarge a specific diagram for closer inspection.

Visualizing Semantic Links: The Interactive Layer also provides visual feedback for the multimodal linking. When a segment of the transcript is being played, any semantically related text on the

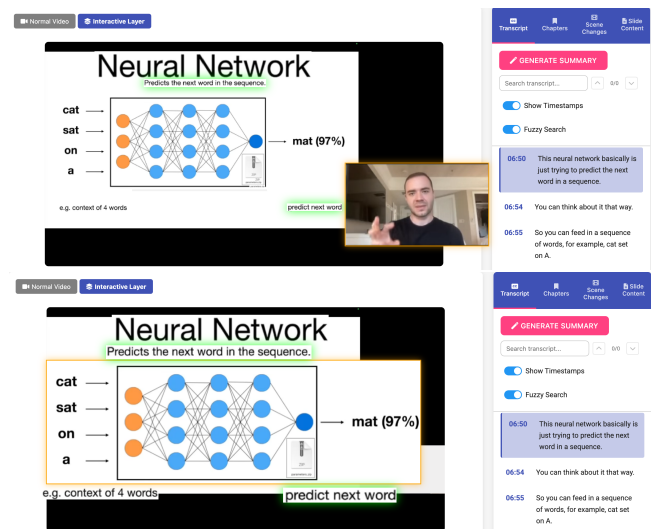


Figure 4: Two versions of a video at the same timestamp. The interactive layer allows user to customize elements size and position. They can also remove elements such as the presenter.

current slide is highlighted with a glow. This works bidirectionally: clicking on a text box on a slide will highlight the corresponding sentences in the transcript panel. This makes the implicit connections between spoken and visual information explicit, reinforcing comprehension.

Text Toggle, Styling and Translation: In addition to layout control, the interface allows OCR-detected slide text to be toggled from its image form into selectable, editable text. Users can then adjust font (including dyslexia-friendly options), color, and size. Rendering text natively instead of as an image also enables on-the-fly translation, as seen in Figure 5, improving accessibility for multilingual learners.

3.2 Personalization as a design driver

A core principle of the toolkit is providing users with a high degree of control over their viewing experience. Early, informal discussions with potential users highlighted that the conventional playback of slide-based lectures can often feel cluttered or difficult to navigate. This feedback was central in establishing three key design goals for our system: providing predictable structure through clear navigational aids, enabling layout personalization to reduce visual distraction, and use multimodal engagement by explicitly linking spoken and written content. These principles guided the development of the specific features detailed above.

4 Application scenarios

The toolkit's features are designed to support several key use cases for students and researchers. For students, the toolkit acts as an interactive study aid. Users can search for keywords across both the spoken lecture and slide content and instantly jump to relevant sections for efficient review. To further support comprehension,

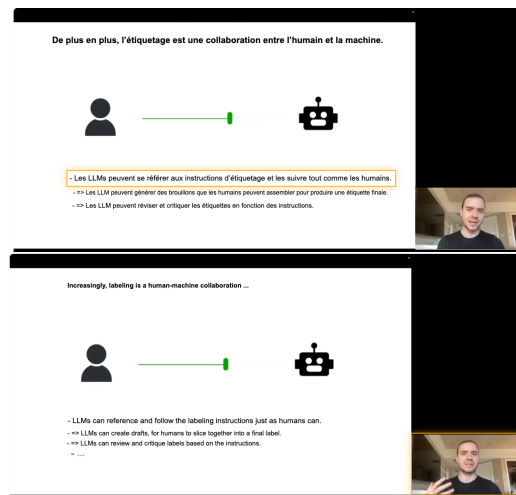


Figure 5: Two versions of a video with different translations. Top: French, Bottom: English.

the interface displays the video, a visual timeline of slides, and a synchronized transcript, with clear, visible links between what is being said and what is shown.

For users who find standard video playback distracting, the 'Interactive Layer' offers customization options. Viewers can personally tailor the layout by hiding the presenter's video, resizing text, or reorganizing slide elements to create a more focused learning environment. Additionally, the toolkit supports on-the-fly translation of on-screen text, making complex material more approachable for a global audience.

Finally, for multimedia researchers, the toolkit serves as an open and extensible platform for experimentation. Its modular architecture allows individual components—such as the OCR engine or embedding model—to be easily modified or replaced. This design facilitates the testing of new algorithms for video analysis and interaction, while its open-source nature supports reproducibility and the extension of prior work.

5 Software availability and deployment

The Video Lecture Analysis Toolkit is released as a fully open-source project. The complete source code is available on GitHub <https://github.com/travisseng/svla-toolkit> with documentation and usage instructions.

The toolkit runs as a backend service. Once deployed, users can upload a video through the web interface or via API endpoints to process lecture content automatically. Output includes segmented transcripts, slide text, speaker-video alignment, and other metadata that can be used for indexing, summarization, or further analysis.

A live demo is available at: <https://travis-seng.fr/svla>. Additional documentation is provided in the repository.

6 Conclusion and Future Work

This paper presented the Video Lecture Analysis Toolkit, an open-source prototype that integrates multimodal analysis techniques

into an interactive system focused on user control and personalization. By combining scene detection, OCR, transcription, and semantic linking, our toolkit provides flexible and meaningful ways for users to navigate, explore, and understand video lecture content. Its open-source nature is a key contribution, offering a platform that researchers and educators can extend and adapt.

The current system has limitations. The accuracy of the pipeline is dependent on the input video quality, and the multimodal linking may require domain-specific tuning. Most importantly, the toolkit requires formal user evaluation to validate its impact on learning outcomes and accessibility.

Future work will address these points and explore several key directions. We plan to enhance accessibility by conducting co-design sessions with neurodivergent users to refine the interface and add new features like audio feedback and automatic layouts. We will also expand the system's event detection capabilities to identify a richer set of events, such as presenter gestures, blackboard writing, pointer movements, and transitions to embedded videos or live annotations. Finally, we aim to broaden the toolkit's applicability by adapting it to handle non-slide-based formats, such as whiteboard and blackboard lectures, which pose unique challenges for segmentation and text extraction.

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